SOP.0024 REV 1.00

xTool Setup and Operation

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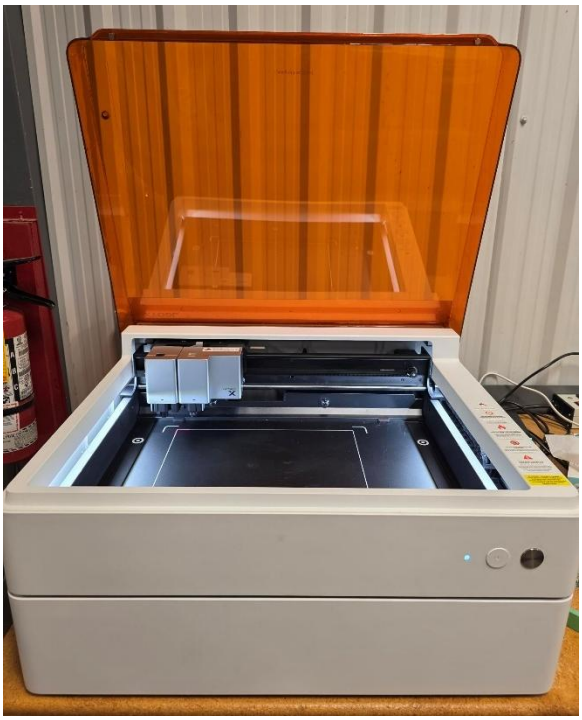
Introduction

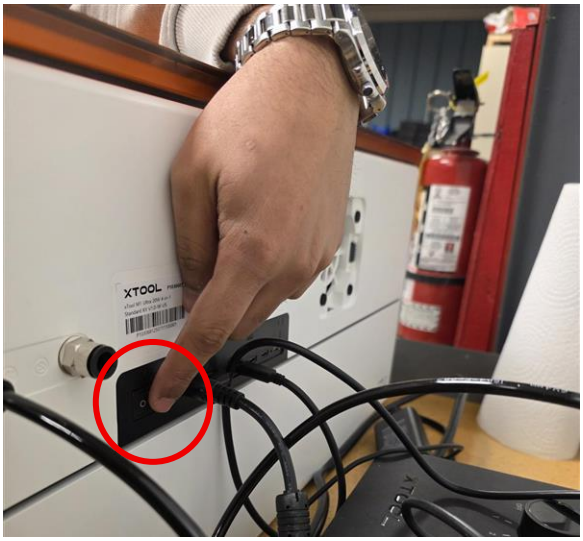
This document provides instructions for operating xTool. The procedure is consistent for all projects, although different materials may be used for the cutouts. The objective is to produce multiple small cutouts using xTool from a single cork sheet and foam sheet.

1 Operating xTool

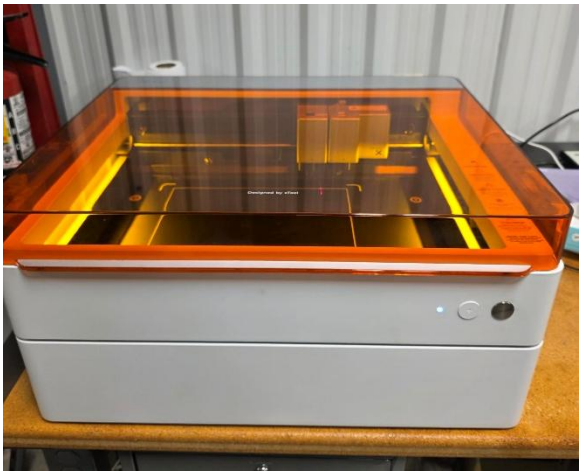


1.1 Setting up xTool

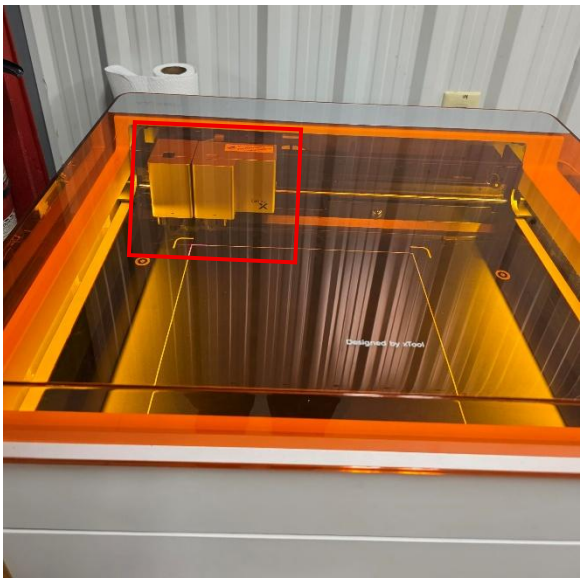
1. Turn on the power switch of xTool.



2. The lights within xTool should turn on, and the laser head should begin repositioning.



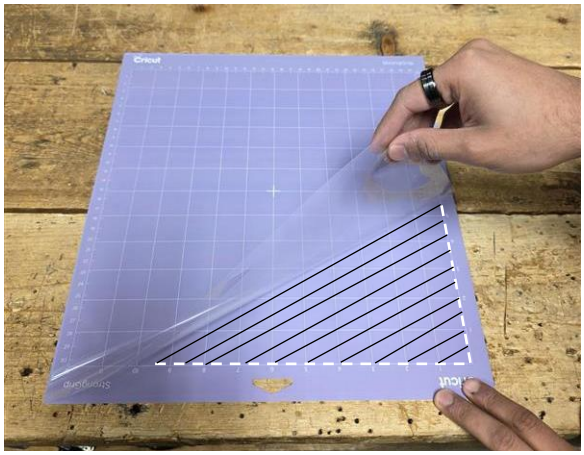
3. Wait until the laser head positions itself at the top-left corner, as shown in the image.



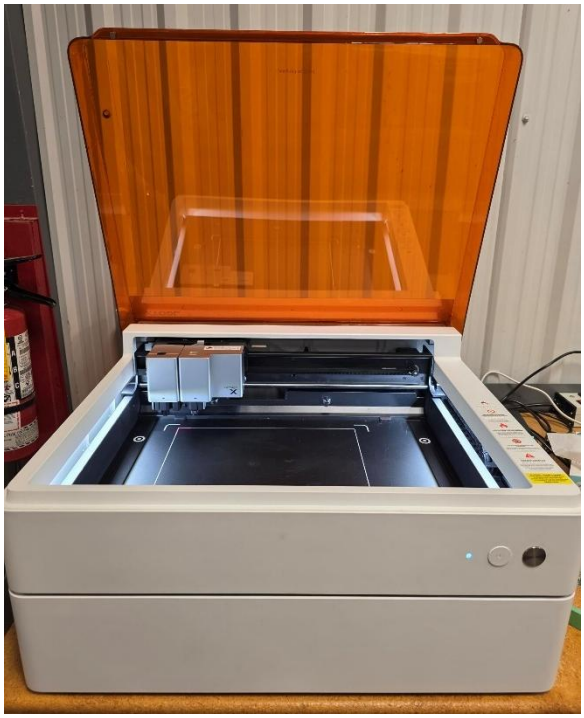
1.2 Preparing Cutting Sheet

1. Prepare a 12 × 12 inch purple-colored cutting sheet for the project. Peel off the outer protective plastic film to expose the **sticky surface**, which appears darker than the regular side.

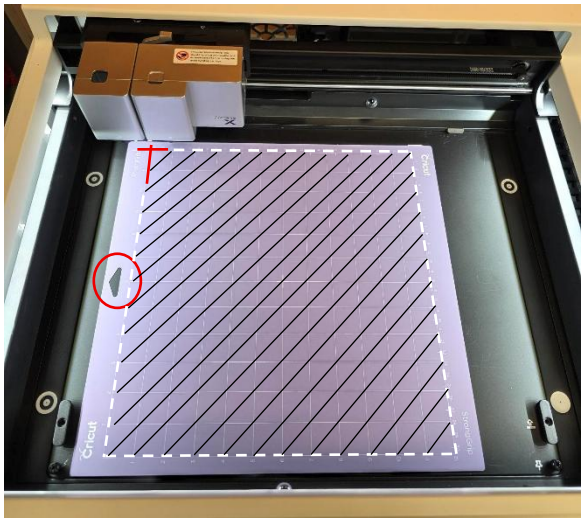
   Note: Set aside the removed protective film to cover the cutting sheet after the project is completed.



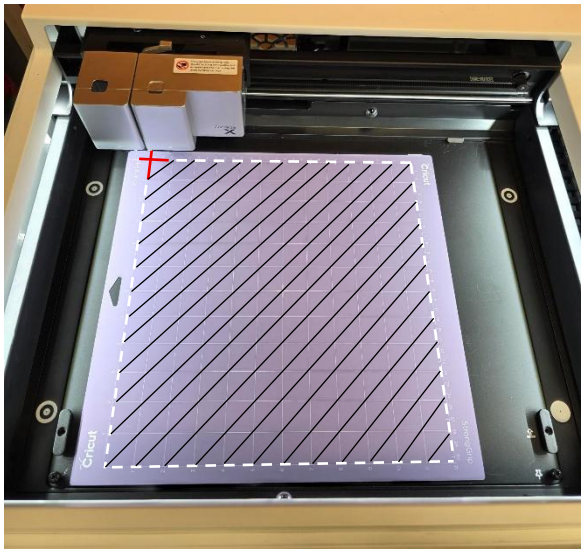
2. Open the lid of xTool.



3. Following the arrow direction, place the cutting sheet inside xTool.

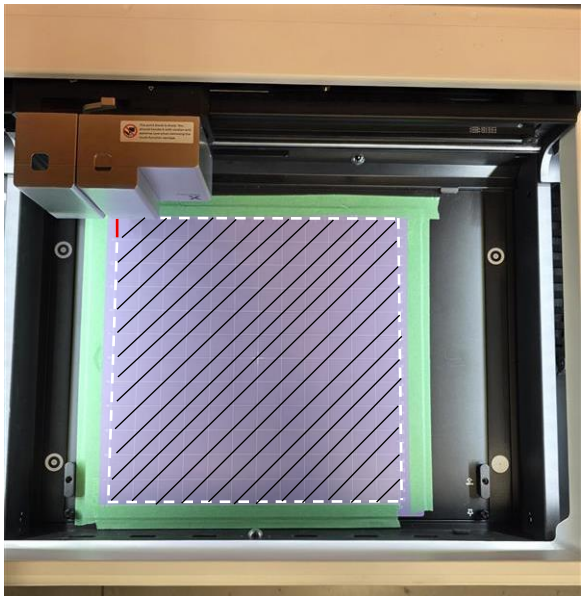


4. Ensure that the laser is aligned with the starting point of the sheet's sticky surface. Multiple attempts may be required to achieve proper alignment.



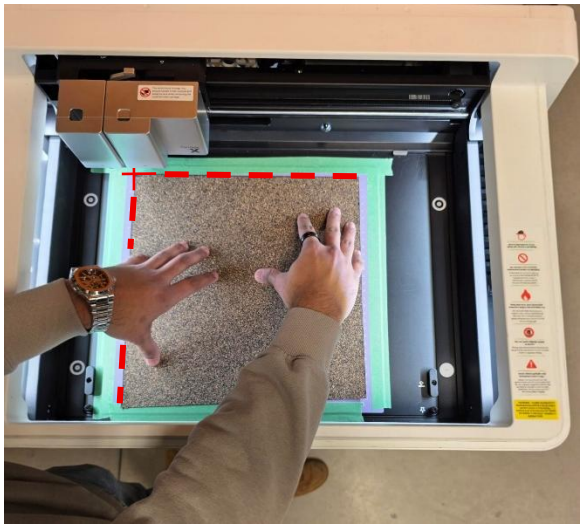
5. Secure the cutting sheet by applying masking tape on all sides, as shown in the image, ensuring that the masking tape does not overlap the sticky surface.

   Note: After this step, you can use either of the following sheets in xTool, one at a time:
   1. **Cork Sheet**: To use a cork sheet, refer to sections 1.3, 1.5.1 and 2.1.
   2. **Foam Sheet**: To use a foam sheet, refer to sections 1.4, 1.5.2 and 2.2.

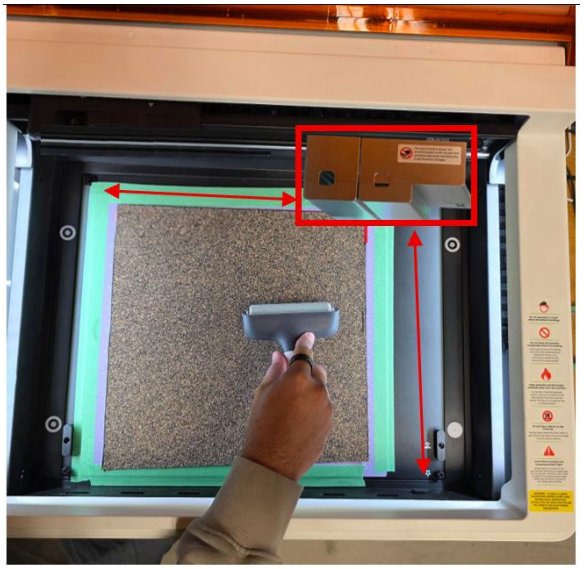


1.3 Using Cork Sheet

1. Place the cork sheet on the sticky surface of the cutting sheet, ensuring that the laser is aligned with the starting point of the cork sheet. Multiple attempts may be required to achieve proper alignment.



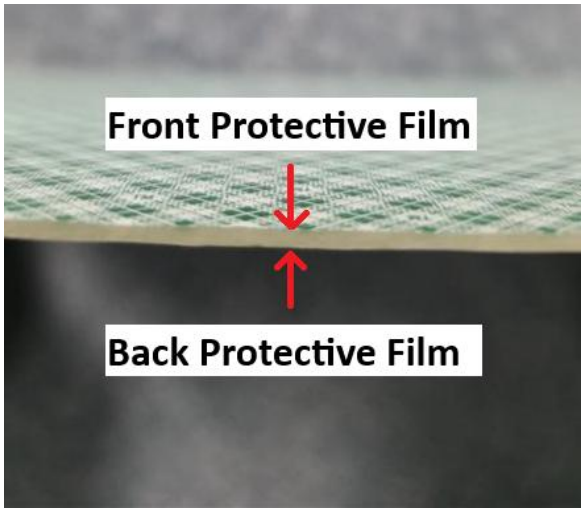
2. Move the laser head to the top-right corner (away from the operator's motion) as indicated in the image.
3. Using the roller, gently roll down the material, ensuring that it adheres to the cutting sheet evenly. There should be no air gaps present.



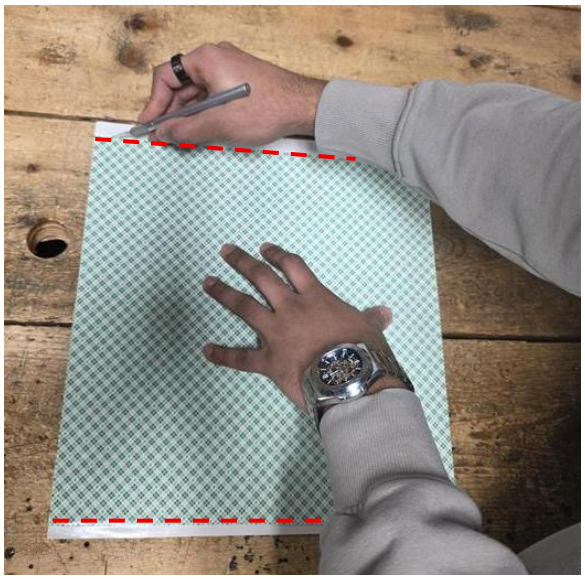
4. Close the lid of xTool.

1.4 Using Foam Sheet

1. The foam sheet is covered with two protective films, as shown in the image.



2. Using an X-Acto knife, trim both edges as highlighted in the image.

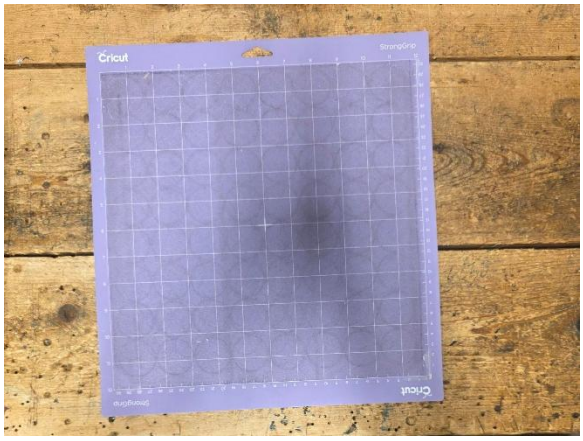


3. Flip the foam sheet.

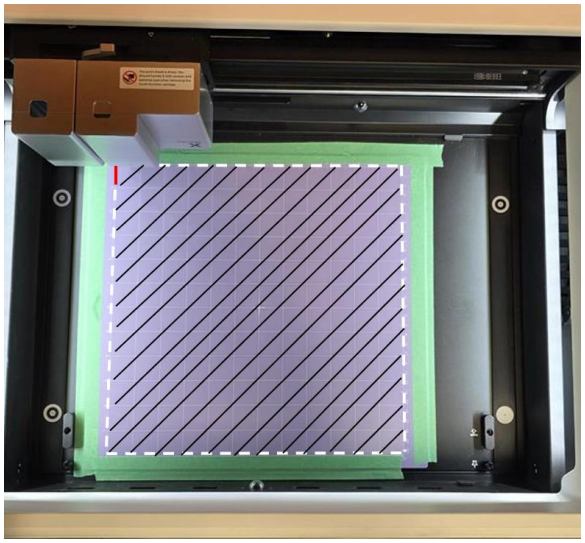
4. Now, gently cut the triangles at all corners, as highlighted. Only cut the back protective film, not the entire foam sheet.

5. Peel off the trimmed protective film. The foam sheet will be exposed at the corners, as highlighted in the image. The foam sheet contains adhesive that sticks to the cutting sheet.

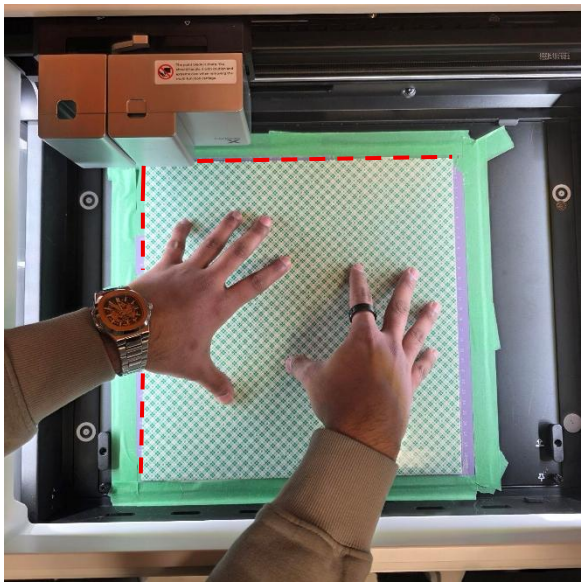
6. A reused cutting sheet can be used with the foam sheet.



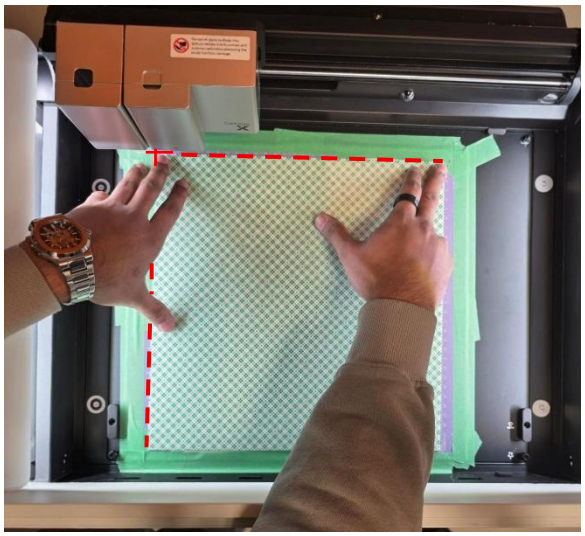
7. Secure the reused cutting sheet by applying masking tape on all sides, as shown in the image, ensuring that the masking tape does not overlap the sticky surface.



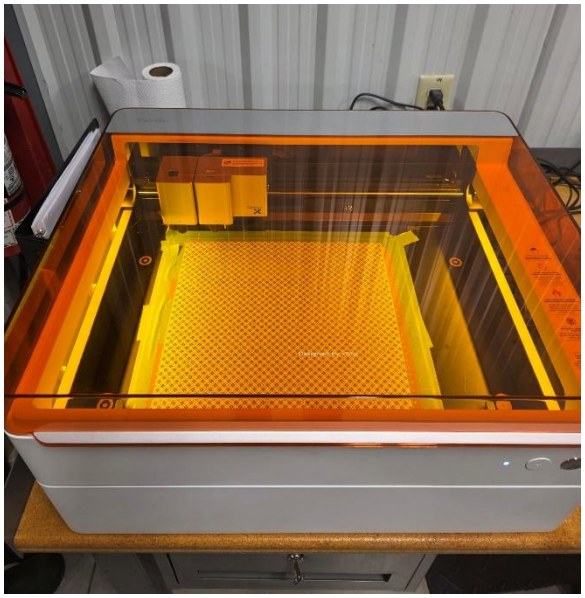
8. Place the foam sheet on the sticky surface of the cutting sheet, ensuring that the laser is aligned with the starting point of the foam sheet. Multiple attempts may be required to achieve proper alignment.



9. Press the corners of the foam sheet so that they adhere to the cutting sheet firmly.



10. Close the lid of xTool.

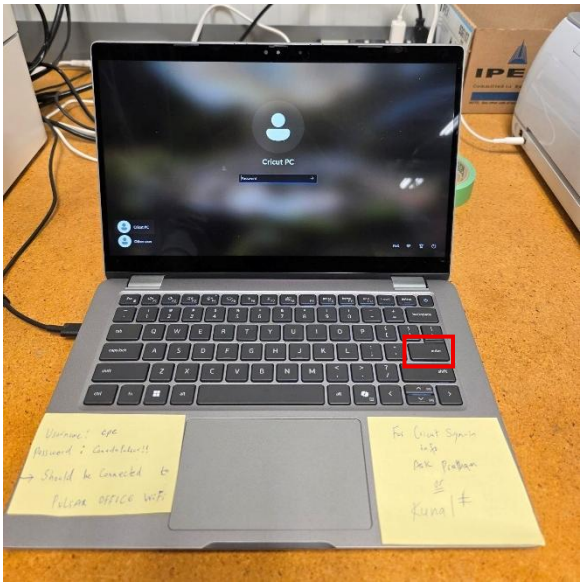


1.5 Configuring xTool Studio Application

1. On the laptop beside xTool, press the power button to turn it on.

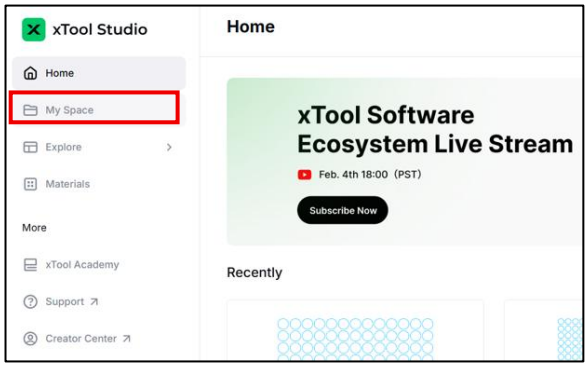
2. When the **Cricut PC** user profile appears, enter the password and press the **Enter** key to log in to the laptop.

   Note: Ensure that the laptop is connected to the Pulsar-Office Wi-Fi network.



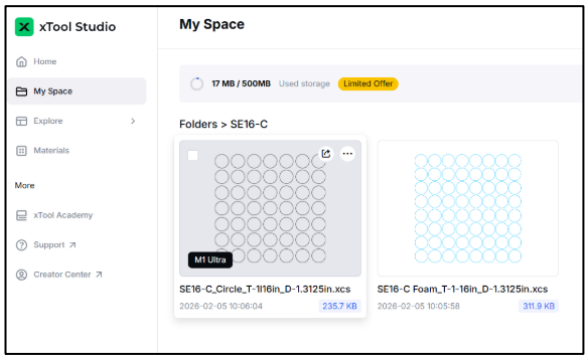
3. On the desktop, open **xTool Studio**. The application may take a few moments to load.

4. Click on **My Space**.

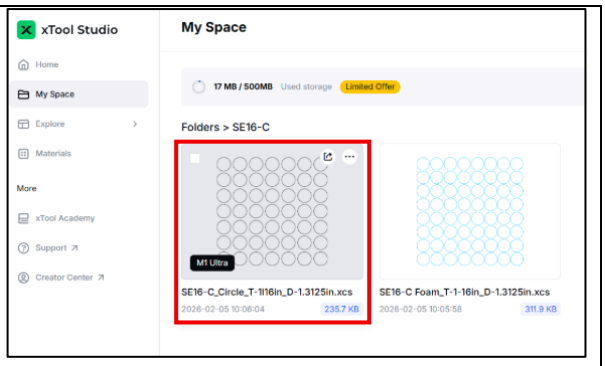
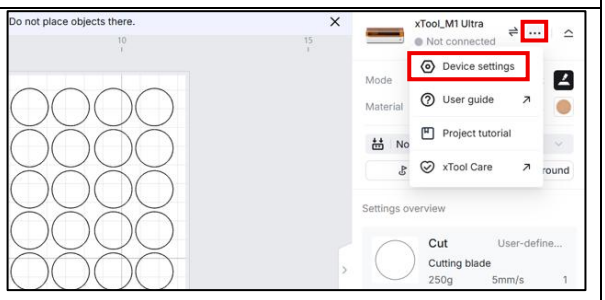
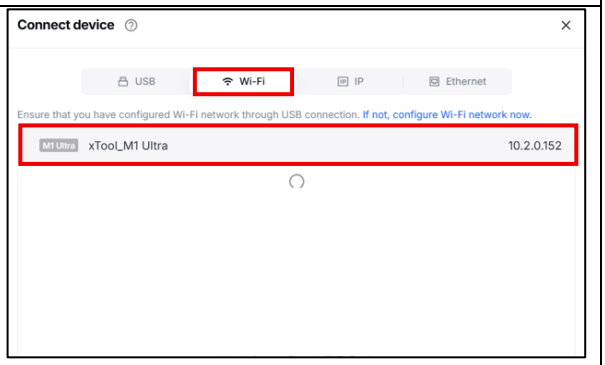
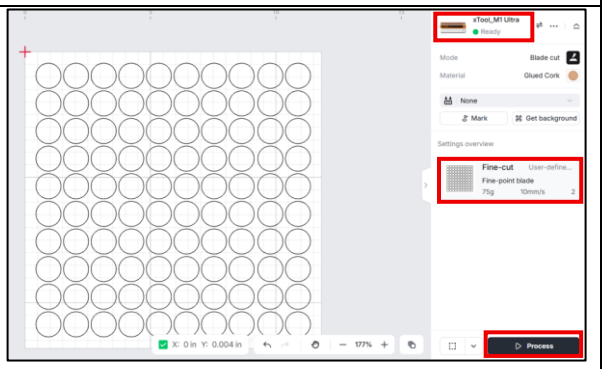


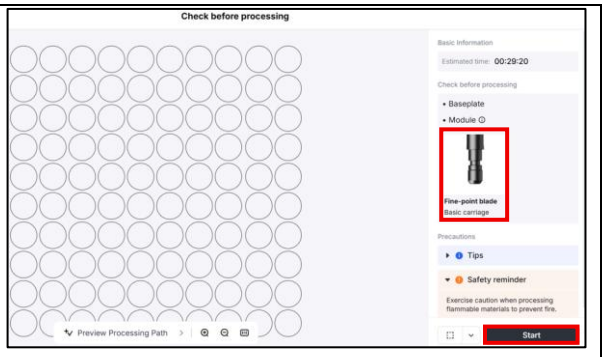
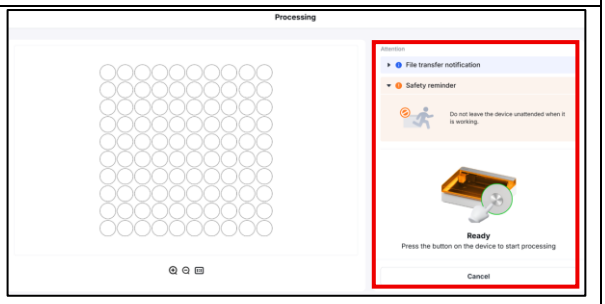
5. The **Folders** section appears. It contains folders with sensor names. Open the appropriate folder.

6. After opening the required folder, select the appropriate file. Each folder contains multiple files, and the material can be identified by the color shown in the cutout preview, as illustrated in the image:
    a. Cork sheet will have cutouts in black Color.
    b. Foam sheet will have cutouts in blue Color.



1.5.1 Processing Cork Sheet

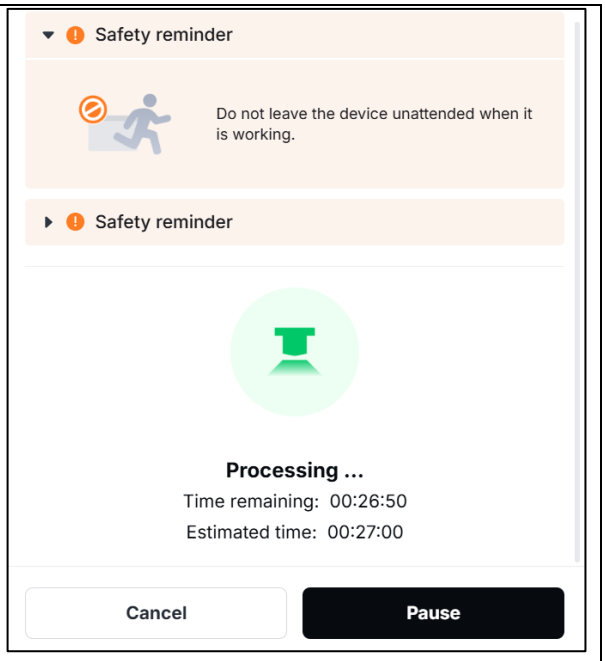
1. Select the appropriate cork sheet file.	
2. After opening the appropriate file, navigate to the top-right corner of the screen and click on the three dots (**...**) beside **xTool_M1 Ultra**. 3. Then select **Device settings** from the drop-down menu.	
4. The **Connect device** window will appear. Click the **Wi-Fi** tab and select **xTool_M1 Ultra**.	
5. The status of **xTool_M1 Ultra** should now display **Ready**. The laser point (+ crosshair mark) should appear at the top-left corner of the canvas. The crosshair shows the real-time position of the laser head. Click the **Process** button in the bottom-right corner of the screen to begin the process. **Note**: The **Settings overview** section shows the type of blade used, which is **Fine-point blade** in this case. Ensure that the same blade usage is displayed on the next screen before starting the process.	

6. On the **Check before processing** screen, verify that the blade type matches the one shown on the previous screen (for example, **Fine-point blade** as shown in the image). Then, click **Start**.	
7. On the **Processing** screen, a message will appear prompting you to start the device for processing.	
8. Press the silver button on xTool to start processing, as shown in the image.	

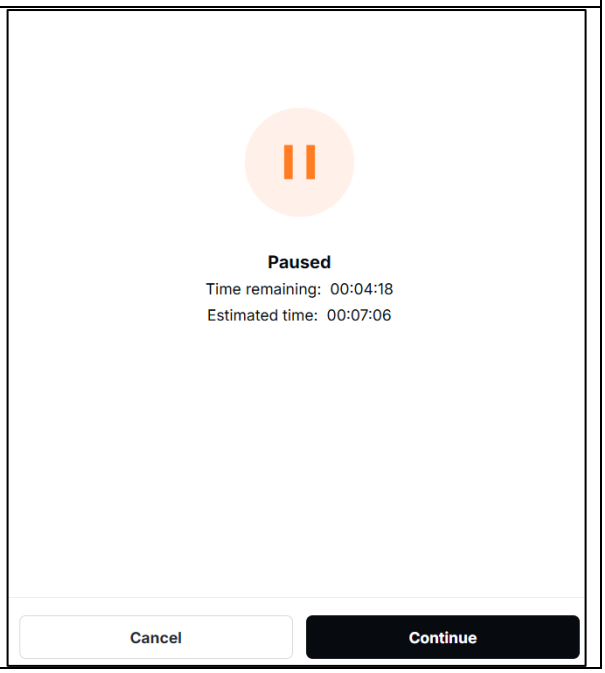
9. Return to the laptop. The estimated processing time for xTool will be displayed.

   Note: While xTool is processing, follow these precautionary steps:
   a. Do not leave xTool unattended.
   b. Do not open the lid of xTool.
   c. Do not press any buttons on xTool, as this may interrupt the process.

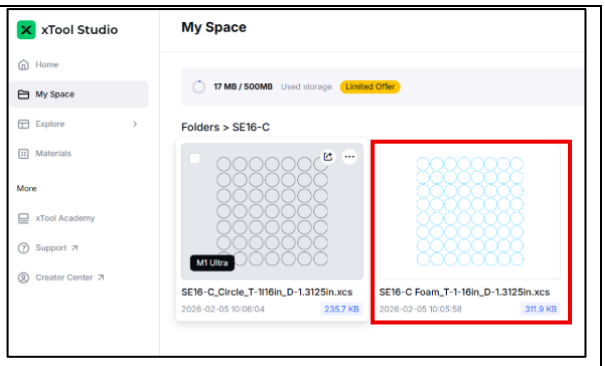
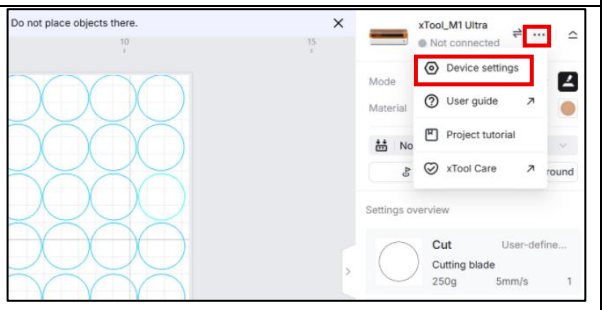
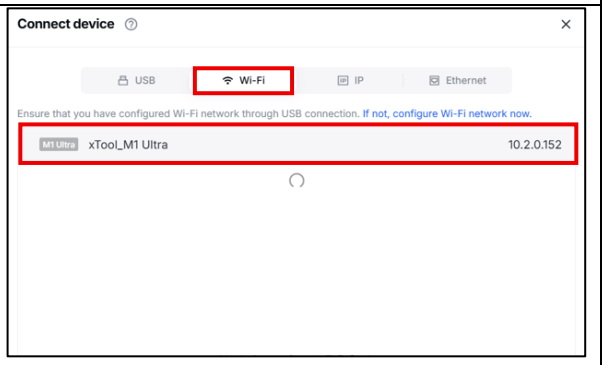
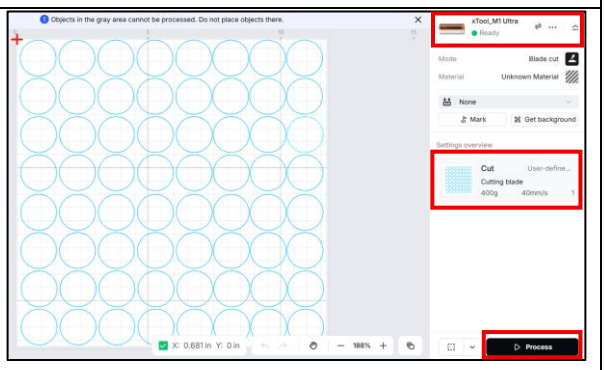
   ⚠ **Warning:**
   In case of an emergency only, press the silver button on xTool or the **Pause** button on the xTool Studio application to stop processing.

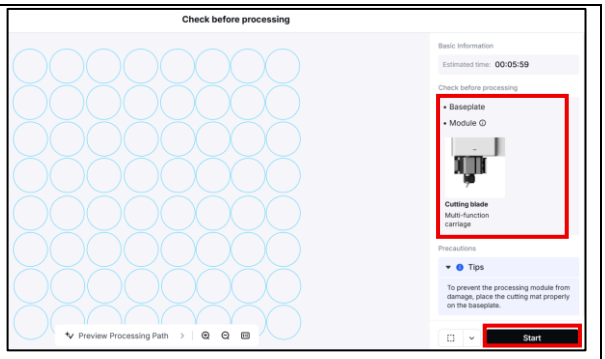
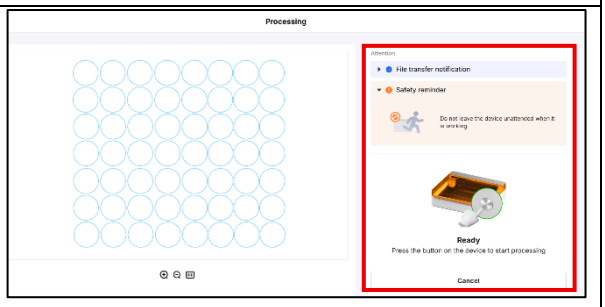


10. Click the **Continue** button to resume the processing of xTool.



1.5.2 Processing Foam Sheet

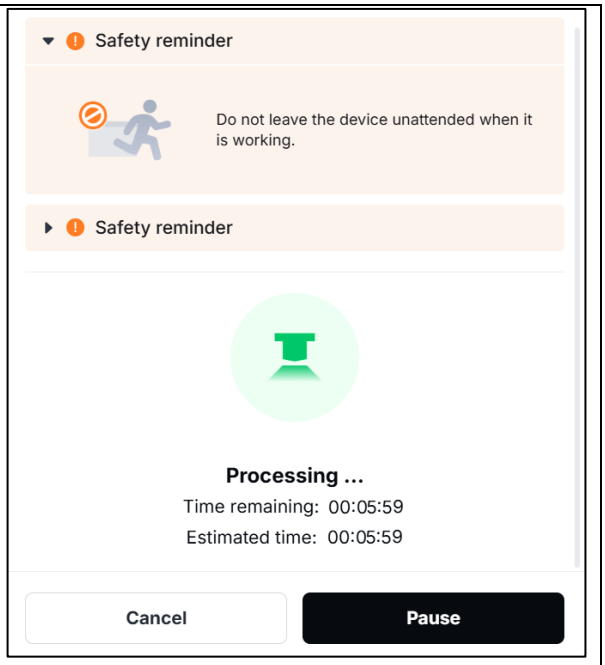
1. Select the appropriate foam sheet file.	
2. After opening the appropriate file, navigate to the top-right corner of the screen and click on the three dots (**...**) beside **xTool_M1 Ultra**. 3. Then select **Device settings** from the drop-down menu.	
4. The **Connect device** window will appear. Click the **Wi-Fi** tab and select **xTool_M1 Ultra**.	
5. The status of **xTool_M1 Ultra** should now display **Ready**. The laser point (+ crosshair mark) should appear at the top-left corner of the canvas. The crosshair shows the real-time position of the laser head. Click the **Process** button in the bottom-right corner of the screen to begin the process. **Note**: The **Settings overview** section shows the type of blade used, which is **Cutting blade** in this case. Ensure that the same blade usage is displayed on the next screen before starting the process.	

6. On the **Check before processing** screen, verify that the blade type matches the one shown on the previous screen (for example, **Cutting blade** as shown in the image). Then, click **Start**.	
7. On the **Processing** screen, a message will appear prompting you to start the device for processing.	
8. Press the silver button on xTool to start processing, as shown in the image.	

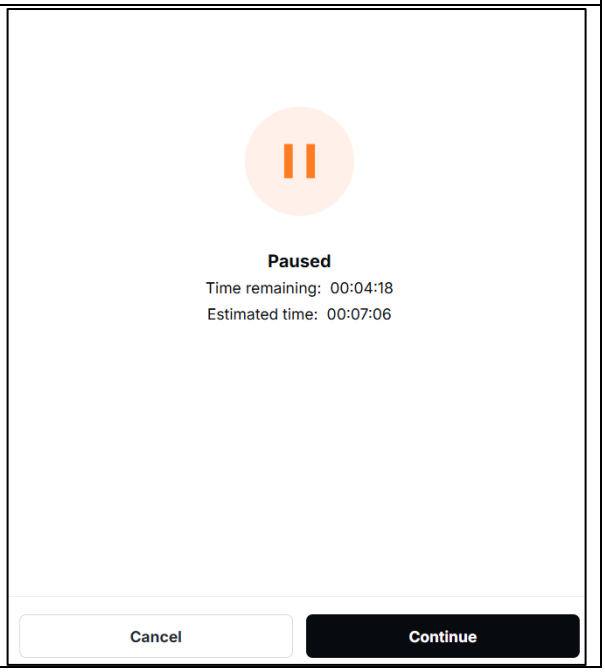
9. Return to the laptop. The estimated processing time for xTool will be displayed.

   Note: While xTool is processing, follow these precautionary steps:
   d. Do not leave xTool unattended.
   e. Do not open the lid of xTool.
   f. Do not press any buttons on xTool, as this may interrupt the process.

> ⚠ **Warning:**
> In case of an emergency, press the silver button on xTool or the **Pause** button on the xTool Studio application to stop processing.



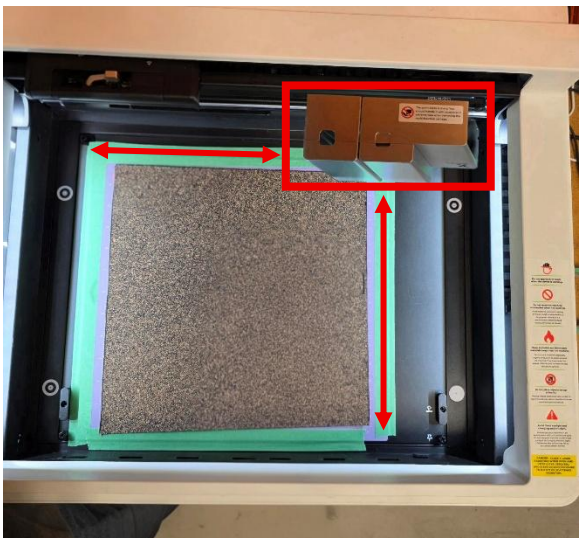
10. Click the **Continue** button to resume the processing of xTool.



2 Getting Cutouts

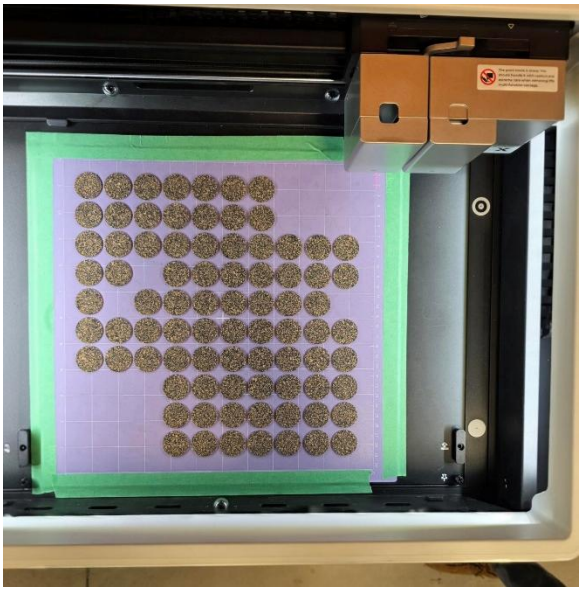
2.1 Securing Cork Sheet Cutouts

1. Once processing is complete, xTool beeps. Open the lid and move the laser head to top-right corner as shown in the image.

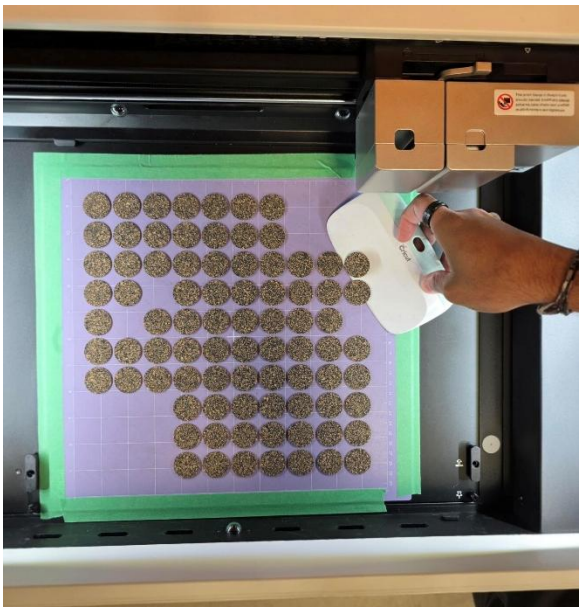


2. Gently start peeling the cork sheet from the top-left corner.

3. Some cutouts will remain attached to the cutting sheet itself, while others may stick to the cork sheet, as shown in the image.

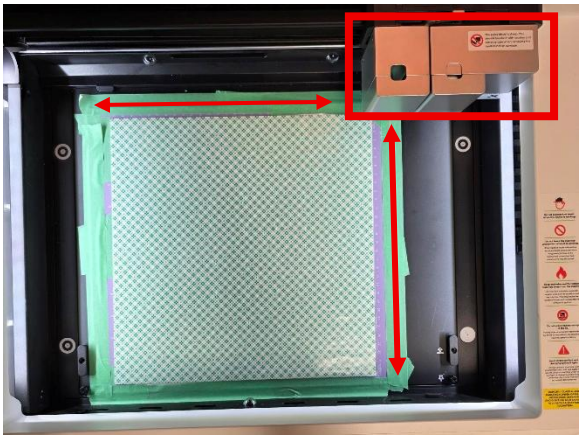


4. Use the cutting sheet scraper to remove all the cutouts from the cutting sheet.

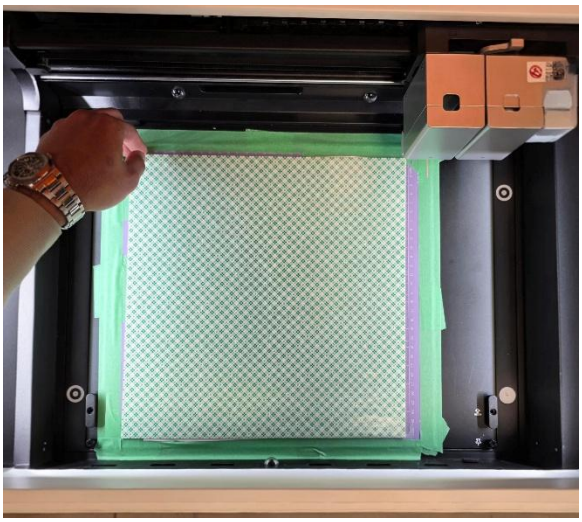


2.2 Securing Foam Sheet Cutouts

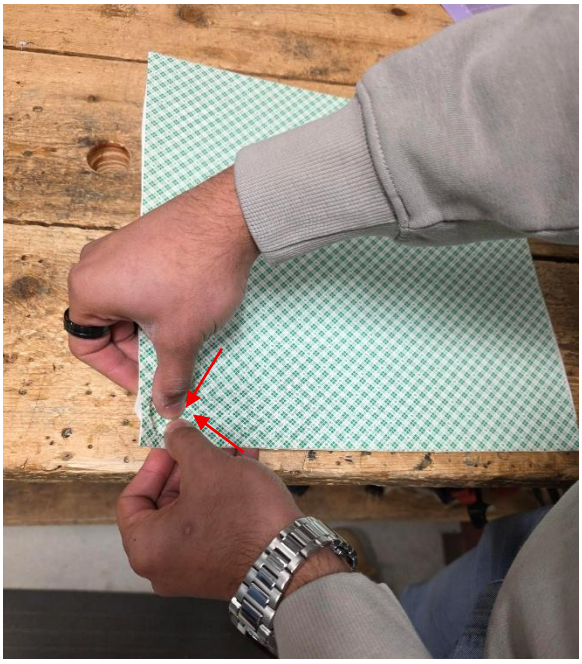
1. Once processing is complete, xTool beeps. Open the lid and move the laser head to top-right corner as shown in the image.



2. Gently start peeling the foam sheet from the top-left corner.



3. Push the cutouts outward so they separate from the foam sheet.



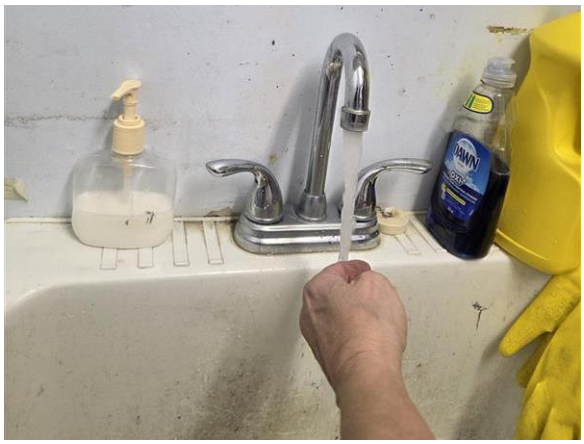
4. Continue to remove all the cutouts from the foam sheet.

3 Cleaning Cutting Sheet

1. After all the cutouts for the day are completed, wash the cutting sheet. To do so, peel off the masking tape and remove the cutting sheet.

2. Close the lid and turn off the power switch of xTool.

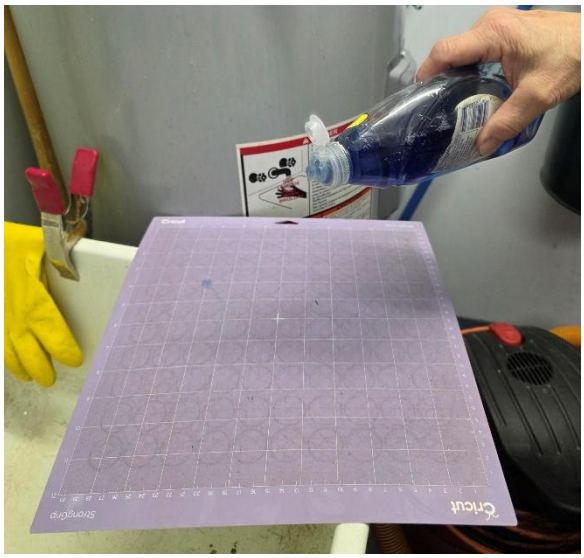
3. Take the cutting sheet to the utility room. Turn on the tap and adjust the water temperature to lukewarm or room temperature.



4. Rinse the front (sticky) side of the sheet first.

5. Then, rinse the back side of the cutting sheet.

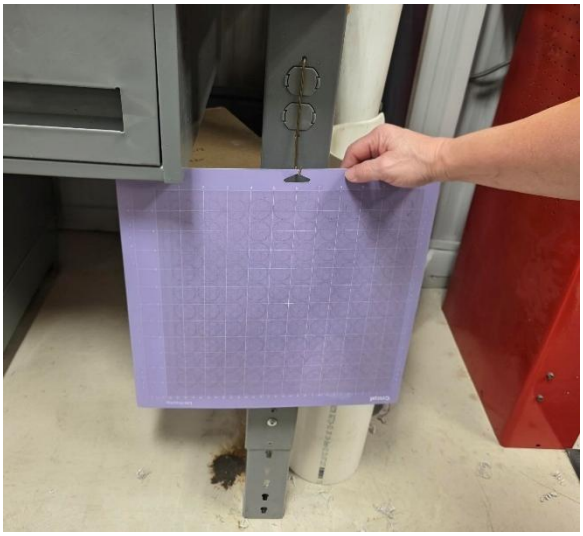
6. Apply a few drops of dish soap onto the sticky surface of the cutting sheet.



7. Hold the cutting sheet in one hand and use the other hand to spread the dish soap evenly over the sticky surface. Gently rub the soap in a circular motion across the entire sheet for a few minutes, taking care not to scrape off the adhesive.

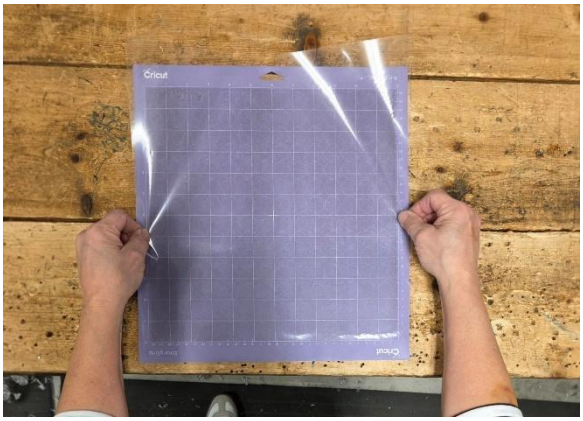
8. Rinse the cutting sheet until all the dish soap is completely washed off.

9. Return to the Machine Shop/Annex and hang the cutting sheet on one of the steel hangers attached to the side of the xTool table. Allow the cutting sheet to drip-dry until the water is dried out completely.



10. Once the cutting sheet is dry, place the protective plastic film over to protect it from dust and dirt accumulation when not in use.

    Note: The cutting sheet must be cleaned once, at the end of the day.



4 Revision Change History

Date	Rev	Summary of Change	Name
February 17, 2026	1.00	First release of the document	Akash Patel